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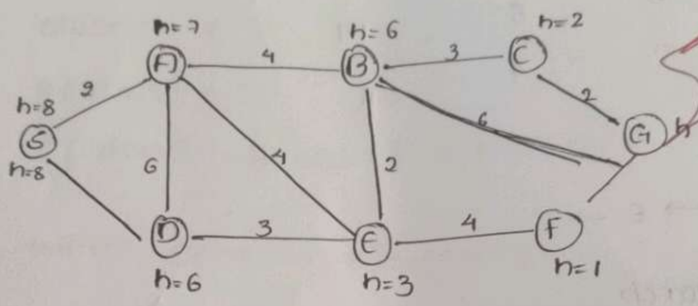
CO2 - AT - 02

PROBLEM SOLVING ASSIGNMENT

Smart Ambulance Navigation



Ans:-

$$f(n) = g(n) + h(n)$$

i) $S \longrightarrow A$

$g(n) = 2$

$h(n) = 7$

$f(n) = 9$

ii) $S \longrightarrow D$

$g(n) = 6$

$h(n) = 6$

$f(n) = 12$

Expand A

from A:

Node	g(n)	h(n)	f(n)
B	6	6	12
D	8	6	14
E	6	3	9

The smallest value is E.

so,

$S \rightarrow A \rightarrow E$

Expand E :

from E

Node	g(n)	h(n)	f(n)
B	8	6	14
F	10	1	11

chose F

$S \rightarrow A \rightarrow E \rightarrow F$

continue the search

$f(C) = 9 + 2 = 11.$

final route:

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Total cost:

$2 + 4 + 3 + 2 = 11$

onclusion:

The ambulance should travel through

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

A* is appropriate because it considers both the cost already travelled.

A* pseudocode.

A* (start, goal)

Open = {start}

closed = { }

g(start) = 0

f (start) = g(start) + h (start)

While Open is not empty:

n = node in open having minimum f(n)

if n == goal :

return path

remove n from Open

add n to closed

< g(m):

parent (m) = n

g(m) = new-g.

return failure.

(2) Autonomous Warehouse Robot :

	1	2	3	4	5	6	7
5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G

Proposed Algorithm: Online A^* / $LRTA^*$

The robot can use an online informed search strat[egy]

such as $LRTA^*$ (Learning Real-Time A^*)

for a grid:

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

Given grid:

(2,4)
(5,4)
(7,3)
(2,2)
(6,2)
(3,1)

The robot should not assume that all unknown are free

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Example Decision Process

S = (1, 5)

All possible sequence after discovering obstacles is :

(1, 5) ⟶ (2, 5) ⟶ (3, 5) ⟶ (4, 5)

It can continue:

(4, 5) ⟶ (4, 4) → (4, 3) ⟶ (5, 3) ⟶ (5, 2) → (5, 1)

Thus, one valid obstacle-free route after the required discoveries is :

S ⟶ (2, 5) ⟶ (3, 5) ⟶ (4, 5) ⟶ (4, 4) ⟶ (4, 3)

- If X is discovered to contain an obstacle :

Planned cell
↓
Obstacle discovered
↓
Mark cell as Blocked
↓
Update environment tool
↓
calculate new heuristic/path
↓
select another cell
↓
Continue toward G.

Conclusion :

An online informed search strategy is appropri[illegible] because the warehouse environment is initially unkn[illegible] the robot can discover obstacles, update its internal and replan dynamically.

③ University Examination Scheduling

Examination	E_1	E_2	E_3	E_4	E_5	E_6
Number of Students	100	80	70	60	90	50

ⓐ CSP Formulation :-

$X = \{E_1, E_2, E_3, E_4, E_5, E_6\}$

Domains

$D(E_1) = D(E_2) = \ldots\ldots = D(E_6)$

$= \{T1, T2, T3, T4\}$

Student Conflict constraints

$E_1 \neq E_2$

$E1 \neq E_3$

$E2 \neq E4$

$E3 \neq E5$

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E4 ≠ E5

E2 ≠ E5

Hall Constraints

Exam	Students	Possible Hall
E1	100	A
E2	80	A/B
E3	70	A/B
E4	60	A/B/C
E5	90	A
E6	50	A/B/C.

(b) MRV + Forward checking

Assignment 1

Exam	Remaining Domain
E1	T2, T3, T4
E2	T1, T2, T3, T4
E3	T2, T3, T4
E4	T2, T3, T4
E5	T1, T2, T3, T4
E6	T1, T2, T3, T4

Assignment 2

$E1 = T2$

$D(E3) = \{T1, T3, T4\}$

Final Timetable

Time slot	HALL A	HALL B	HALL C
T1	E3 - 70	E2 - 80	-
T2	E1 - 100	E4 - 60	-
T3	E5 - 90	-	E6 - 50
T4	-	-	-

GAME - PLAYING AI :

The supplied game tree has a MAX root, thr MIN nodes, and MAX nodes below them.

Step 1 :

MAX nodes :

MAX (3, 5, 6) = 6

MAX (9, 1, 2) = 9

MAX (0, -1, 4) = 4

MIN chooses:

MIN (6, 9, 4) = 4

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Therefore, Left branch = 4

Middle MIN branch

MAX (7, 5, 8) = 8

MAX (6, 2, 3) = 6

MAX (4, 0, -2) = 4

MIN (8, 6, 4) = 4

Right MIN branch.

MAX (1, 2, 3) = 3

MAX (5, -1, 2) = 5

MAX (6, 4, 7) = 7

MIN (3, 5, 7) = 3

Root MAX

MAX (4, 4, 3) = 4

Final Minimax Decision

The AI should select left or middle move, because both have a minimax value of :

4

Alpha - Beta pruning.

$\alpha = 4$

MAX (1, 2, 3) = 3

$\beta = 3$

$\beta = 3 < \alpha = 4$

5, -1, 2, 6, 4, 7

So, 5 terminal values can be avoided in this left to right evaluation.

5 Delivery Route Optimization - Hill Climbing

Solution	Route	Cost
S1	D → A → C → B → E → D	26 Km
S2	D → B → A → C → E → D	30 Km
S3	D → A → B → E → C → D	24 Km
S4	D → A → E → B → C → B	27 Km
S5	D → E → C → B → A → D	23 Km
S6	D → C → B → A → E → D	29Km

initial neighbouring costs:

26, 30, 24, 27, 23, 29

The smallest is:

23 Km

Ex: 17

55 is selected.

step : 2

The assignment explicitly starts from 55.

23 → 22

since no neighbouring solution has a cost lower than 22 km, Hill climbing stops.

Final cost

22 km

How to Escape a local optimum :

Genarate random route.

↓

Appy Hill climbing

↓

Reach local optimum

↓

Restart with another route

↓

Compare solutions

Conclusion:

The Hill climbing process improves the initial solution from the provided neighbourhood, selecting S5 with 23 km, and then moves to a neighbour with 22 km. Since all neighbours of the 22-km solution have cost at least 22 km, the algorithm stops at a local optimum.